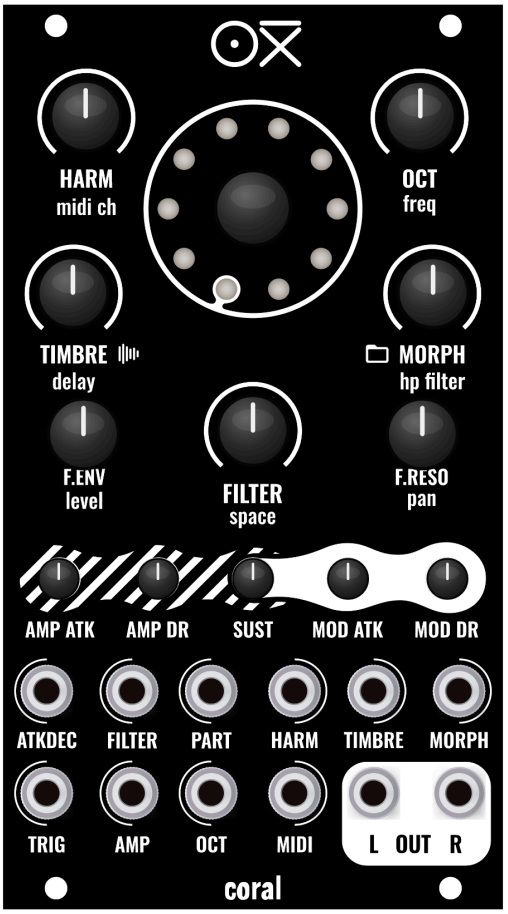


USER GUIDE

V 3.0

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WELCOME

Thank you for purchasing OXI Coral.

The introduction of the Mutable Instruments Plaits module was a revolution in compact, multi-engine synth voices. Coral builds on those open-source foundations. It packs 8 independent voices into a 14 HP module. Each voice has its own envelopes, filter and effects. The interface is colourful and organic. On top of that you get custom wavetables and a sample playback engine per voice.

But Coral is not just another polyphonic synth. The multipart concept lets you arrange the 8 voices however you like: 8 monophonic parts, one 8-voice polyphonic part, or any combination. A single Coral can be a polysynth, a drum machine, a groovebox and a sound-exploration module at the same time.

Coral firmware 3.0 introduces a new way of building your instrument. Instead of a handful of fixed engine sets, there are now 27 engines and you choose any 10 of them. The web Firmware Updater compiles that exact selection and flashes it to your Coral over USB. This manual covers that workflow, the new engines, the on-device options menu, and the companion web apps.

The OXI team

HOW TO USE THIS MANUAL

This guide follows the signal from your fingers to the output. Chapters 1 to 4 cover the hardware, the interface and how to play Coral. Chapter 5 documents every engine. Chapters 6 to 8 cover effects, MIDI and the SD card. Chapter 9 covers the companion web apps. Chapter 10 is calibration.

If you are brand new: read Chapter 1, plug in MIDI or a Gate/CV pair, and start playing. Coral behaves as an 8-voice polyphonic synth out of the box.

- **NOTE** boxes add useful context.
- **CAUTION** boxes flag things that can interrupt sound or lose data.
- **TIP** boxes are shortcuts and good habits.

CONTENTS

1 OVERVIEW

Coral is a polyphonic, multipart synthesizer voice module for Eurorack. It is polyphonic: it plays up to 8 notes at once. It is multipart: each of the 8 voices runs its own engine, listens to its own MIDI channel, and holds its own parameter values.

The simplest way to play Coral is as an 8-voice polyphonic MIDI synth. You can also play it as a single voice from CV. That is how it behaves the first time you power it on. To hear how Coral sounds, patch a Gate and a CV/Oct pair. These trigger the internal envelopes.

1.1 Specifications

Dimensions	14 HP, 3 cm deep
Panel	Black anodized aluminium with colour silkprint
Consumption	+12V: 110 mA · -12V: 10 mA · +5V: 0 mA
Outputs	2x mono DC-coupled outputs (±5 V), used as a stereo pair
Inputs	1x MIDI TRS (Type A and B compatible) · 8x CV inputs ±5 V (±10 V max) · 1x Gate input
Storage	Micro SD card, always in the slot. Class 10 / 70 MB/s or faster for heavy sampler use

1.2 Panel Overview

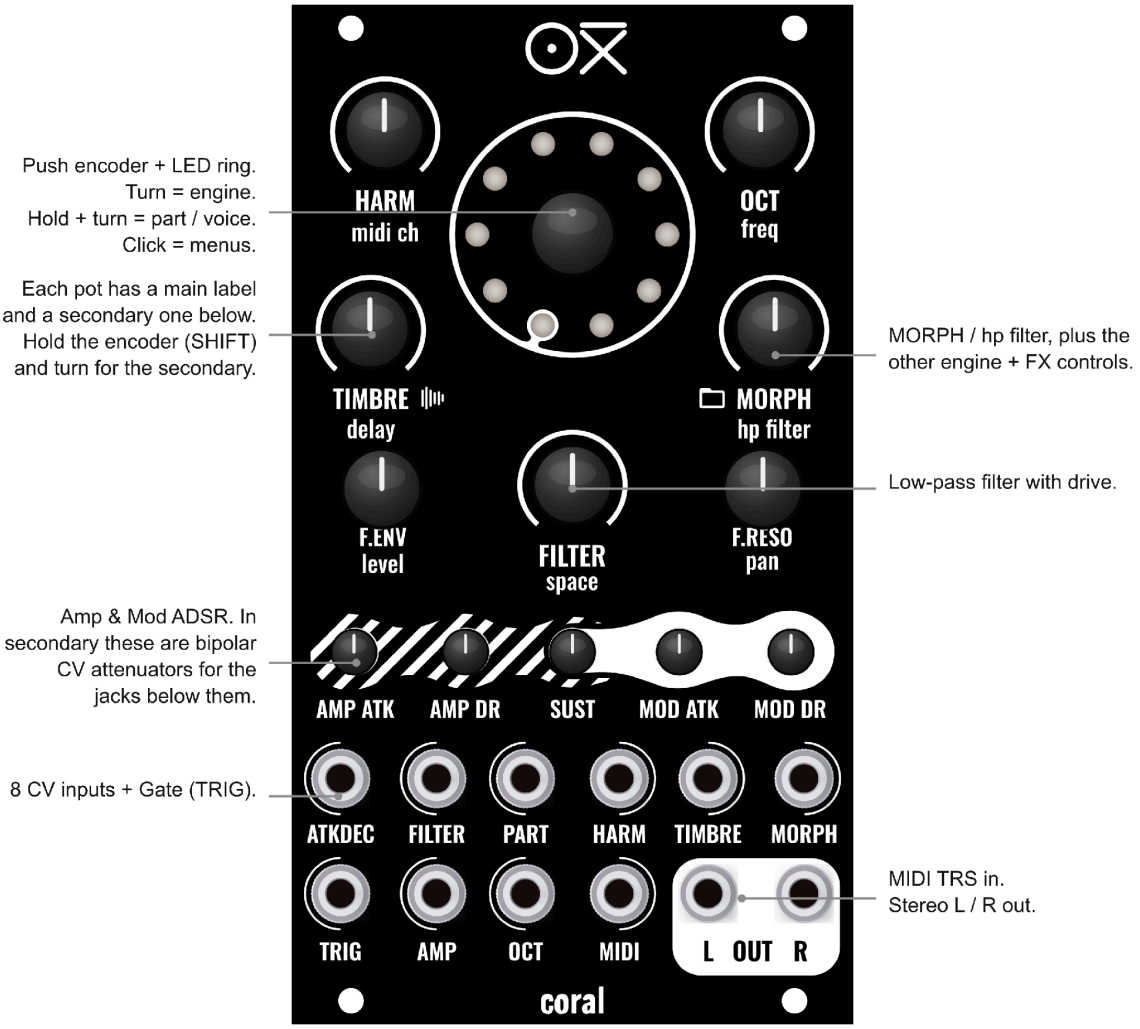


Figure 1.1. The Coral panel. Push encoder and LED ring. A main label and a secondary label on each pot. HARM, OCT, TIMBRE and MORPH pots. LP filter with drive. Amp and Mod ADSR controls, which also work as CV attenuators. SD card slot. 8 CV inputs and Gate. MIDI input. Stereo L and R outputs.

- **Encoder with push button.** Turn to select the engine. Push and hold, then turn, for voice / part selection. Push and hold plus turn a pot to reach that pot's secondary parameter. Click to step through the LOAD, SAVE, CV routing and options screens (Chapter 2).
- **Micro SD card.** Always keep it in the slot. You need a fast card (Class 10 or higher) for heavy sampler use.
- **CV inputs.** Affect the selected part (all voices by default). A part can be addressed with the Part CV jack or in the CV Assign screen. The ADSR trimmers double as bipolar CV attenuators in their secondary function.
- **LED ring.** Visual feedback for every screen. By default each LED shows the amplitude of one voice.

1.3 Powering Up

1. Make sure the factory Micro SD card is inserted.
2. Mount Coral in a powered Eurorack case with the ribbon cable oriented correctly (red stripe to -12V).
3. Power on. The LED ring shows voice activity in the main view.
4. Send notes on MIDI channel 1. You can also patch a Gate to the trigger input and a CV to the OCT input. Then play.

> **NOTE** By default all 8 voices listen to MIDI channel 1 and are assigned as one part. This is the best starting point.

1.4 Signal Flow

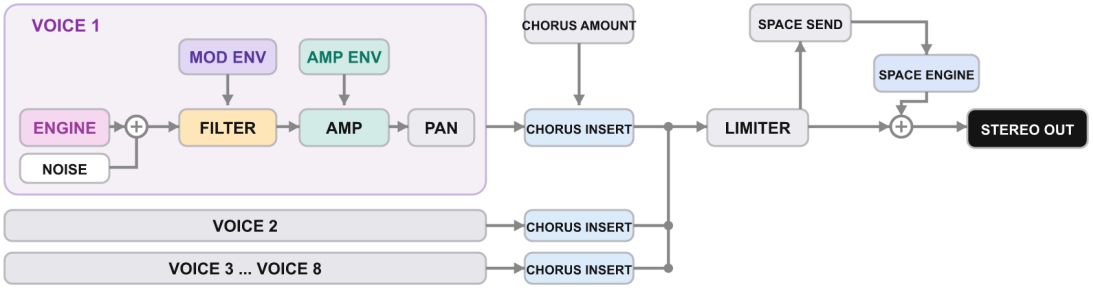


Figure 1.2. Per-voice signal chain. Engine and Noise go into the Filter, then the Amp, then Pan. The Mod and Amp envelopes shape the Filter and the Amp. Each voice has its own Chorus insert. The Limiter sums all 8 voices. The Space reverb is a send and return, then the stereo output.

Each voice runs its engine and an independent noise source through a low-pass filter and an amplifier. The Mod and Amp envelopes shape it. Coral then pans it. The chorus works per voice. A limiter sums all voices. The Space (reverb) engine is a shared send. The result reaches the stereo output.

2 INTERFACE AND PLAYING

This chapter covers the main view and the pots, which have two functions each. It also covers engine selection, the encoder click cycle, presets, the options menu and the randomizer.

2.1 Main View

In the main view the LED ring shows the amplitude of each active voice. This is the default screen; turning any pot returns you here from any menu.

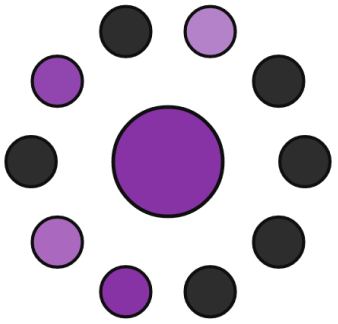


Figure 2.1. Main view, voice status: voices 1, 2, 4 and 6 active.

2.2 Primary and Secondary Parameters

Each pot controls one main parameter and one secondary parameter. This lets Coral pack a lot of control into a small panel. To change a secondary parameter, push and hold the encoder (this acts as SHIFT) and turn the corresponding pot.

The pot position and the stored value may not match, so to avoid sudden jumps you first have to *catch up* the stored value. This applies to both main and secondary parameters. For a primary parameter the stored value is shown in the section colour and the pot position in white. For a secondary parameter the stored value is white and the current position is orange.

> **NOTE** At start-up the pot position and the stored value may not match. You may then need to catch up the main or secondary values.

Taking control back from MIDI

The same idea applies when something else is driving Coral. A sequencer or DAW can move a parameter with a MIDI CC. The pot then no longer matches the sound. The pot must catch up before it takes control again. As you turn, the pot and the value meet in the middle. Control passes smoothly, without a jump. The same happens when you switch part or load a preset.

It applies to every parameter a CC can reach, on both layers:

- **Main functions:** HARM, TIMBRE, MORPH, FILTER cutoff, F.RESO, F.ENV, AMP ATK, AMP DR, MOD ATK, MOD DR and SUST.
- **Secondary functions:** Level, Pan, Noise, Space (reverb) send, Delay send, and Chorus send or HP cutoff.

An incoming CC now arms catch-up properly, including the Delay send, which it did not before. If you would rather the pots stayed live at all times, set **noccxt 1** in options.txt (Chapter 8). That suits a

sequencer streaming CC automation or parameter locks, where constant re-arming would otherwise make a pot feel dead.

> **NOTE** Recalling an engine is the one exception. Coral restores a per-engine sound as described in Section 2.4. The pot must reach the exact remembered position. The sound then lands precisely on what you saved.

> **NOTE** HARM and OCT behave differently in their secondary functions, which are MIDI channel and fine tune. They take over the moment the pot crosses the stored value. They do not slew towards it. This is how they have always worked.

2.3 The Controls

Primary and secondary functions affect the selected voice, or the voices in the selected part, only. HARM, TIMBRE and MORPH are the three engine parameters and do something different in every engine (see Chapter 5).

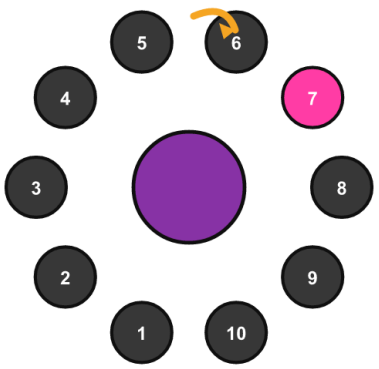
Control	Main function	Secondary (hold encoder + turn)
ENCODER	Turn: select engine. Click: feature select (sample / bank / wavetable) then LOAD, SAVE, CV, options. Hold + turn: part / voice.	None
HARM	Engine Parameter 1	MIDI channel
OCT	Base octave (offset of the MIDI note or CV/Oct pitch)	Fine tune
TIMBRE	Engine Parameter 2	Noise level (Delay send if Delay is on)
MORPH	Engine Parameter 3	HP filter cutoff (if HP on) or Chorus send
F.ENV	Mod-envelope to filter amount, bipolar (0% at 12 o'clock)	Level
FILTER	Low-pass filter cutoff	Space (reverb) send
F.RESO	Filter resonance and drive	Pan (constant-power)
AMP ATK	Amp-envelope attack	Atk/Dec CV attenuator
AMP DR	Amp-envelope decay and release	Filter CV attenuator
SUST	Sustain (amp and mod envelopes)	Harm CV attenuator
MOD ATK	Mod-envelope attack	Timbre CV attenuator
MOD DR	Mod-envelope decay and release	Morph CV attenuator

Noise. The noise level (SHIFT + TIMBRE) feeds an internal white-noise source. That source is independent of the voice level, so you can layer noise freely. The MOD (modulation) envelope drives the filter cutoff by default. The MOD controls can also change the sound, depending on the engine you select.

> **NOTE** A few engines re-use these controls for something else. The STITCHER engine in particular puts its main control on OCT and its loop levels on HARM, TIMBRE and MORPH. See Section 5.8.

2.4 Engine Selection

Turn the encoder to change the engine of the selected voice or part. The new engine is applied immediately. The active engine of the selected part is shown in pink on the ring.



Turn to select the engine. Active slot lit pink.

Figure 2.2. Engine-selection ring: 10 engine slots around the ring, the active slot lit pink, remaining slots dim. Slot contents depend on the firmware you built.

> **NOTE** Your 10 engine slots are whatever you chose in the Firmware Updater (Chapter 9). The example rings in this manual are just one possible layout.

What happens to your sound when you switch engine

Every engine on a part keeps its own tone settings. Move to another engine and come back. That engine returns exactly as you left it. It keeps HARM, TIMBRE and MORPH, the filter cutoff, the filter envelope and resonance, both envelopes, sustain and noise.

Your effects mix does not move. The Space (reverb), Delay and Chorus sends belong to the part, not to the engine. The HP filter cutoff does too. Set a reverb and delay balance you like. You can then change engines on stage. The mix stays put and the delay keeps running.

> **TIP** This is what makes engine switching usable on stage. Set the send levels once for the part. You can then move through engines without the effects jumping or dropping out.

2.5 The Encoder Click Cycle

Clicking the encoder steps through a series of screens. The LED ring colour tells you where you are. Turning a pot at any time exits back to the main view.

Ring colour	What this stop does
Lime / amber / orange	Feature stop. It appears only on engines that have something to pick. FMX7 picks a DX7 bank. WAVETABLE and WT SWARM pick a wavetable. DUAL WAVETABLE picks Table A, then Table B. STITCHER picks a sample, and has a second stop for its FX ring. Every other engine has no feature stop, so its first click is LOAD.
Blue	LOAD preset. Turn to pick a slot, click to load.
Red	SAVE preset. Turn to pick a slot, click to store.
Purple	CV routing. Choose which voices the Part CV jack steers (only when Part CV is unpatched).
Options ring	The options menu (Section 2.7). Hold about 1 second to exit to playing.

Order: feature stop (if any), Blue LOAD, Red SAVE, Purple CV, options ring, then back to playing. A green sweep confirms an action; red means reset.

2.6 Presets

You can save and load all parameters and the full voice configuration in 10 preset slots. The extended-presets option gives you 100 (Chapter 8). Everything is stored, so a preset recalls exactly as you left it. Presets can also be recalled with MIDI Program Change messages.

Tap the encoder to jump between screens. Turning the encoder in any screen enters selection mode, so the next tap confirms. Turn any pot to exit to the main view.

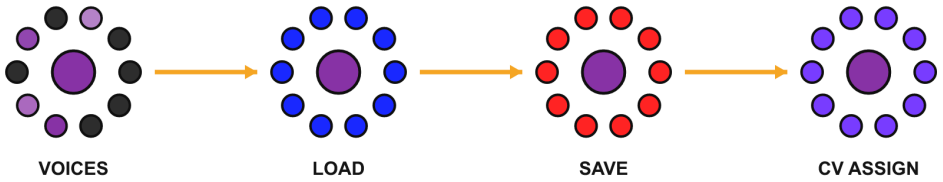


Figure 2.3. Screen navigation strip VOICES to LOAD (blue) to SAVE (red) to CV ASSIGN (purple).

> **TIP** Each saved preset remembers its own wavetable. Older presets fall back to the global default wavetable.

2.7 The Options Menu

The options ring lets you change Coral's performance settings without a computer. Click the encoder past LOAD, SAVE and CV to reach it. Turn to move the orange cursor. A short click toggles a slot or opens a picker sub-view. A slot glowing its own colour is OFF; white means ON. Hold about 1 second to exit, or leave it alone and it closes itself after about 4 seconds.

Coral saves changes straight away. It also writes them into options.txt on the SD card, so the two always agree. You can set the same options in that file instead if you prefer (Chapter 8).

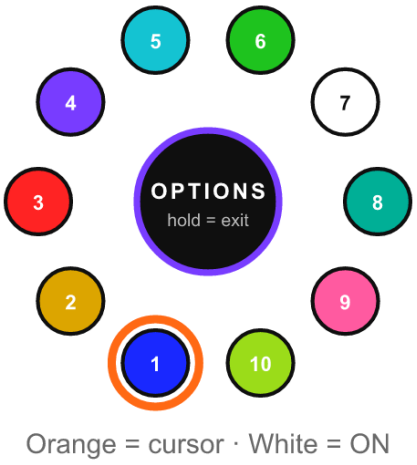


Figure 2.4. Options ring. 10 LED slots sit around the encoder gap. LED 1 is bottom-left and LED 10 is bottom-right. The orange dot is the cursor. One slot is white, which means ON.

LED	Option	What it does
1	**PolyCV**	Voice / trigger mode 0 to 5: unison, poly (CV picks the voice), chord hold, chord arp, then two echo modes. Opens a numbered sub-view.
2	**Encoder part priority**	Encoder turn selects the part / voice instead of the engine.

LED	Option	What it does
3	**CV part to engine**	The Part CV jack selects the engine instead of the part.
4	**MPE**	MPE in and out.
5	**Program Change**	0 accept all, 1 ignore all, 2 accept only on an active part channel. Opens a numbered sub-view.
6	**Randomizer**	Hold the encoder to randomize the sound (Section 2.9).
7	**High-pass filter**	On: SHIFT + MORPH is HP cutoff. Off: SHIFT + MORPH is chorus send.
8	**Delay FX**	On: SHIFT + TIMBRE is delay send. Off: SHIFT + TIMBRE is noise level.
9	**Drone**	Opens the drone picker. Drone keeps the notes you play sounding after you let go of the keys (Section 2.8).
10	**Smart chord**	One-finger diatonic chords, on or off per engine (String 2 and PIANOZ today). The sub-view mirrors your 10 slots and the capable engines light up. The scale is picked separately, on the encoder feature stop.

NOTE The blue, cyan and lime slots are pickers. These are PolyCV, Program Change and Smart chord. Click to open the sub-view. Then click a number or an engine to set it.

2.8 Drone

Drone is a setting of the part you are on. When it is on, the notes you play keep sounding after you let go of the keys.

Turn drone on and off

1. Open the options ring and select LED 9 (Drone). The drone picker opens.
2. Look at the picker. It shows one LED for each of your 10 engine slots.
3. Click a blue engine. The part changes to that engine and starts to drone.
4. To stop the drone, click the white engine again.

A blue LED is an engine that can drone. A white LED is the engine that this part drones on. When you stop the drone, the held notes fade out with the RELEASE knob.

The encoder selection sets which part the drone applies to. On a part, it applies to that part. On a single voice, it applies to the part of that voice. On ALL, it applies to every part.

How a drone plays

- Nothing sounds until you play a note.
- Play a chord and let go of the keys. The chord keeps sounding.
- The next chord you play replaces the previous chord. You always hear the last chord that you played.
- ATTACK shapes the start of each note. After the attack, the level stays at full, whatever DECAY and SUSTAIN are set to.
- A note that leaves the drone fades out with RELEASE. This happens when you play the next chord, or when you stop the drone.
- If you change the engine of a droning part, the drone moves to the new engine.

Each part has one drone. Different parts can drone at the same time. For example, 4 parts of 2 voices each can drone 4 different engines, and each part plays its own 2 notes.

Limits

- The voice caps of the engines still apply: FMX7 has 6 voices, ACID 2 has 4 voices and STITCHER has 1 voice. Notes above the cap do not sound.
- The drum engines (BASSDRUM, SNARE, HI-HAT and SNARE / HI-HAT) and WAVPLAYER cannot drone. If you change a droning part to one of them, the drone stops.
- All Notes Off (MIDI CC 123) stops the droning notes. Drone stays on for the part.
- With Smart chord (STRING 2, PIANOZ) or in MPE mode, the level stays at full while you hold the keys. The notes release when you let go.
- Some engines have their own internal envelopes, for example some FMX7 patches. Drone keeps the Coral VCA open, but it cannot stop the envelope of the engine. These sounds can still fade.

> **TIP** The preset saves the drone setting of each part. A droning pad or texture comes back as you left it.

2.9 Randomizer

Turn on the Randomizer slot in the options ring. Then hold the encoder to randomize the sound. With one part selected, it randomizes that part. With all parts selected, it gives each part its own new patch. It keeps the engine, level and pan.

2.10 Voices and Part Reset

Hold the encoder from the main view for at least 5 seconds. Wait for the red LED animation. Coral returns to its default state: all voices on MIDI channel 1, and all parameters at their init values.

3 MULTIPART AND VOICES

Coral has 8 voices. This chapter shows how to group them into parts. You can then play a mono bassline, a mono lead and a block of polyphonic chords at the same time.

3.1 The Parts Concept

Voices set to the same MIDI channel belong to one part. By default all 8 voices are on MIDI channel 1, so there is a single part. As soon as you assign a voice to a different MIDI channel, a new part is created. Each part can be addressed by its MIDI channel or with the Part CV jack.

Flexibility has a price. We recommend starting with one part. Once comfortable, create two parts and practise switching between them, then build up from there.

How do you know which voice plays a note? Coral assigns a free voice to every MIDI note it receives. If all voices are busy, it steals the least recently used one. The trigger (Gate) input triggers all voices of the selected part.

3.2 Part and Voice Selection

To navigate voices and parts, push, hold and turn the encoder. From the default state, holding the encoder for half a second lights the ring to show all voices selected. Any change then applies to all voices.

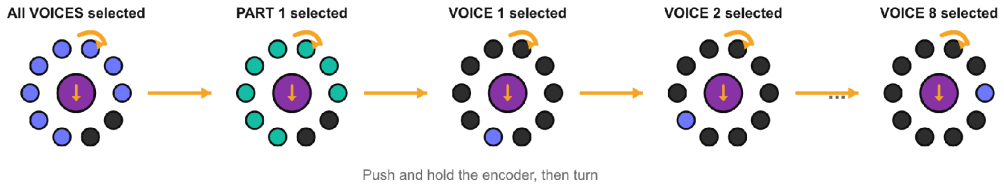


Figure 3.1. Selection sequence: All VOICES selected, PART 1 selected, then VOICE 1, VOICE 2 and on to VOICE 8. Voices occupy LEDs 1 to 8.

- Hold the encoder and turn one tick clockwise to move from All Voices to Part selection.
- Turn another tick to step to voice 1, then voice 2, and so on to voice 8.
- Past voice 8 the selection wraps back to All Voices, then Part selection.
- When more than one part exists, Part selection steps through each part in turn.
- Turning counter-clockwise navigates the other way.

3.3 Creating Parts

This example builds three parts. Voice 1 is a mono bassline on MIDI channel 1. Voice 2 is a mono lead on MIDI channel 2. Voices 3 to 8 are 6-voice polyphonic chords on MIDI channel 3.

1. Set all voices to channel 3 first, so you do not change each voice one by one. Hold the encoder until all voices are selected. Hold the encoder and turn HARM until the ring shows MIDI channel 3. The secondary function of HARM is the MIDI channel. Release the encoder. All voices are now on channel 3, and they form the chords part.
2. Hold the encoder and turn it to voice 1. Hold the encoder and turn HARM until the ring shows MIDI channel 1. Release the encoder. Voice 1 is now its own part.
3. Repeat for voice 2, assigning MIDI channel 2.

During Part selection the three parts now appear. Part 1 is the bassline, Part 2 the lead, and Part 3 the chords and pads.

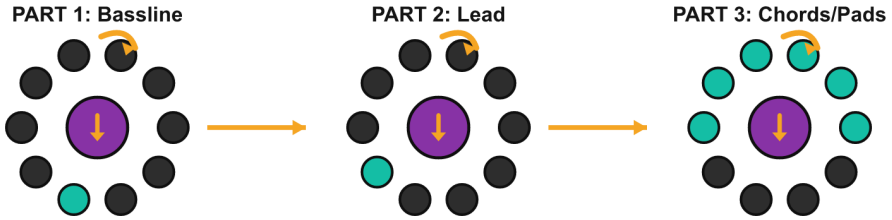


Figure 3.2. The three parts during PART selection: PART 1 Bassline (voice 1), PART 2 Lead (voice 2), PART 3 Chords/Pads (voices 3 to 8).

3.4 Voices and Part Reset

Hold the encoder from the main view for at least 5 seconds. Wait for the red LED animation. This returns all voices to MIDI channel 1 and all parameters to their init values.

4 CV AND MIDI MODULATION

There are two sources of modulation: MIDI (CC and Pitch Bend) and the CV inputs. Coral adds the knob position and the CV input to get the value of a voice parameter. The last CC value replaces the knob position. This applies when a CV is assigned to that voice or part.

- Knob value is unipolar, except the bipolar parameters: Pan, Filter envelope and Frequency.
- CV modulation is bipolar: ±5 V.
- CC is unipolar and overrides the knob value.

Parameter value = knob (or CC) + CV modulation.

Each part can be modulated separately, which lets you build richer, more complex sounds.

4.1 CV Modulation

The Atk/Dec, Filter, Harm, Timbre, Morph, Trig, Amp and Oct CV inputs affect the voices of the selected part only. By default there is one part, so all voices respond.

Poly and Trigger Modes (PolyCV)

The PolyCV setting in the options ring sets how CV and the trigger drive the voices. It has six modes:

Mode	Behaviour
0 · Unison	All voices play in unison.
1 · Poly CV	Different CV pitch values trigger different voices. You can play Coral polyphonically from a single Oct CV and one trigger.
2 · Chord hold	Holds a chord across voices.
3 · Chord arp	Arpeggiates the held chord.
4 · Echo	Echo mode.
5 · Echo	Second echo mode.

NOTE In modes 2 to 5, incoming MIDI notes take over the pitch. You can play chords and echoes from a keyboard as well as from CV.

CV Inputs

Input	Effect
Atk/Dec	Increases or decreases attack, decay and release of both the Amp and Mod envelopes. Negative voltages make the sound pluckier, positive voltages softer and longer.
Filter	Positive voltages raise the filter cutoff, negative lower it.
Harm, Timbre, Morph	Positive voltages apply positive modulation on these engine parameters, negative apply negative.
Trig	Triggers the internal envelopes.
Amp	Controls the amplifier level. It scales every engine.
Oct	CV/Oct pitch. The Oct and Frequency (fine) knobs are taken into account for the final pitch.
Part	The voltage on this input selects which part the other CV inputs affect.

The ADSR trimmers also work as bipolar attenuators for the CV inputs below them. Amp Atk scales the Atk/Dec CV. Amp Dec scales the Filter CV. Sust scales the Harm CV. Mod Atk scales the Timbre CV. Mod Dec scales the Morph CV.

4.2 Part CV Input

One way to choose which part the CV modulation affects is the Part CV jack. Only 0 to +5 V is considered. With two parts, the first is addressed between 0 and 2.5 V and the second between 2.5 and 5 V. There is a little hysteresis to avoid jumps at the edges.

> **NOTE** Turn on the CV part to engine option on the options ring, LED 3. The Part CV then selects the engine instead of the part.

4.3 The CV Assign Screen

Leave the Part CV jack unpatched. You can then send the front-panel CV modulation to one part with the CV Assign screen. This is the purple stop in the encoder click cycle. Turn the encoder to choose the part. Click to confirm. Any voltage on the Part CV jack overrides this selection.

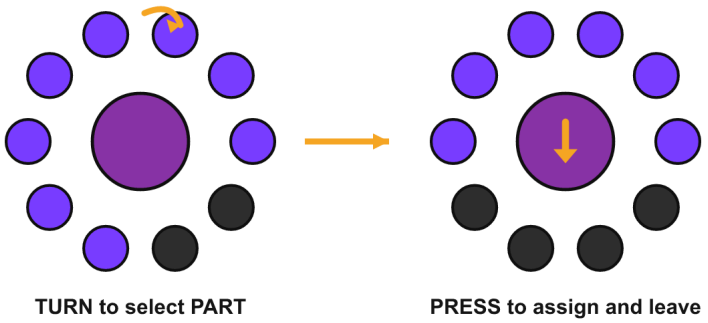


Figure 4.1. CV Assign ring: turn to select the part, press to assign and leave.

4.4 MIDI Modulation

With MIDI CC messages you can modulate any parameter inside Coral. Messages are filtered by MIDI channel and applied to the part on that channel. CC is unipolar and overrides the knob value. See Chapter 7 for the full CC table.

5 ENGINES

An engine is the sound generator inside a voice. Coral firmware 3.0 ships 27 engines. You choose any 10 of them when you build your firmware in the Firmware Updater (Chapter 9). The 10 you pick fill the 10 slots on the engine-selection ring, in the order you set. This chapter documents all 27 so you can decide what to load.

> **NOTE** The sampler (WAVPLAYER) is one of the 27 engines, not a separate mode. Some engines are heavier than others, so not every combination of 10 fits. The Updater shows live Flash and RAM meters. It tells you before you build.

The three engine knobs. HARM, TIMBRE and MORPH are Engine Parameters 1, 2 and 3. What they do changes per engine, and is listed with each engine below. Over MIDI they are CC 11, 12 and 13.

5.1 Analog and Subtractive

VIRTUAL ANALOG

Classic two-oscillator subtractive analog. The second oscillator is synced to the first.

Knob	Parameter	Does
HARM	Engine Parameter 1	Oscillator detune, following musical intervals
TIMBRE	Engine Parameter 2	Square shape, from narrow pulse to full square to hard-sync formants
MORPH	Engine Parameter 3	Saw shape, from triangle to saw with a widening notch

3-VCO ANALOG

Three variable-saw oscillators that morph from triangle to saw to square to pulse with dynamic PWM. Two of the three can be detuned against the main oscillator.

Knob	Parameter	Does
HARM	Engine Parameter 1	Oscillator 2 detune (calibrated to land on intervals such as +7 and +12)
TIMBRE	Engine Parameter 2	Waveform of the oscillators (triangle, saw, square, PWM)
MORPH	Engine Parameter 3	Oscillator 3 detune

SH-101

Monophonic SH-101 voice with a sub-oscillator.

Knob	Parameter	Does
HARM	Engine Parameter 1	Sub tuning
TIMBRE	Engine Parameter 2	Waveform and pulse-width
MORPH	Engine Parameter 3	Sub level

MDO

Multiple Detuned Oscillator: 8 detuned oscillators per voice morphing between saw, square and PWM.

Knob	Parameter	Does
HARM	Engine Parameter 1	Detuning amount

Knob	Parameter	Does
TIMBRE	Engine Parameter 2	Waveform morph, saw to square
MORPH	Engine Parameter 3	PWM of the square waveform

MDO2

A thicker multi-oscillator swarm.

Knob	Parameter	Does
HARM	Engine Parameter 1	Detune spread
TIMBRE	Engine Parameter 2	Waveform
MORPH	Engine Parameter 3	Voice count

OB-VA

Oberheim-style virtual-analog poly voice with a Curtis-style multimode filter and analog drift.

Knob	Parameter	Does
HARM	Engine Parameter 1	Detune / spread
TIMBRE	Engine Parameter 2	VCO combination (upper half of the range = unison)
MORPH	Engine Parameter 3	Filter slope, 4-pole to 2-pole

> **NOTE** Set the obvspr option to 1 for per-voice analog spread (Chapter 8). This gives a subtle drift between voices.

5.2 Waveshaping and Additive

WAVESHAPE

An asymmetric triangle processed by a waveshaper and a wavefolder. Bright and digital.

Knob	Parameter	Does
HARM	Engine Parameter 1	Waveshaper waveform
TIMBRE	Engine Parameter 2	Wavefolder amount
MORPH	Engine Parameter 3	Waveform asymmetry

ADDITIVE

Harmonic / additive synthesis with two kinds of distortion: wavefolding and diode emulation. Ranges from mild tones to heavily distorted basslines.

Knob	Parameter	Does
HARM	Engine Parameter 1	Harmonic content center
TIMBRE	Engine Parameter 2	Wavefolding amount
MORPH	Engine Parameter 3	Diode distortion amount

5.3 FM

2-OP FM

Two sine-wave operators modulating each other’s phase: bells, keys and metallic tones.

Knob	Parameter	Does
HARM	Engine Parameter 1	Frequency ratio
TIMBRE	Engine Parameter 2	Modulation index (FM depth)
MORPH	Engine Parameter 3	Feedback: operator 2 modulating its own phase (past 12 o'clock, rough) or operator 1 (before 12 o'clock, chaotic)

FMX7 (DX7)

A full 6-operator FM engine that loads DX7 .syx banks from the SD card.

Knob	Parameter	Does
HARM	Engine Parameter 1	Patch: turn to step through the patches in the current bank
TIMBRE	Engine Parameter 2	FM brightness
MORPH	Engine Parameter 3	Juno-60 chorus (Mode I+II), a shared post-mix BBD chorus. 0 is dry and CPU-free. Turn it up for the classic wide Juno wobble

Polyphony	Limited to 6 voices (other engines: 8). A 7th note steals the oldest.
Patches	.syx files in the SD folders DX7 Patches/Group 1 to 10. Each folder is one bank.
Change bank	Click the encoder once: the ring lights one lime dot for the current bank. Turn to another bank, click to return to playing. HARM then browses that bank.
Per part	The bank and the patch belong to the part. Several FMX7 parts can run completely different DX7 sounds at the same time. A bank change on one part leaves the others alone.
Saved	The preset stores the bank next to HARM. A DX7 sound needs both to identify it, so both travel with the preset. They also survive a switch to another engine and back.
Slides	Tied / legato notes glide on a mono part. Accent via note velocity.

5.4 Wavetable

WAVETABLE

Scans wavetables loaded from the SD card. HARM, TIMBRE and MORPH move through the Z, X and Y dimensions; three waves are combined and cross-faded.

Knob	Parameter	Does
HARM	Engine Parameter 1	Z axis
TIMBRE	Engine Parameter 2	X axis
MORPH	Engine Parameter 3	Y axis

Files	.wav wavetables (27 x 512 waves, 48 kHz 16-bit) in the SD Wavetables/ folder, up to 96. Make your own in the Waveform Generator (Chapter 9).
Browse	Click the encoder: the first stop (dark-yellow arc) picks the wavetable. Turn to scrub, click to exit.
Memory	The wavetable persists across power cycles, and each saved preset remembers its own.

DUAL WAVETABLE

Two detuned wavetable oscillators for complex lead sounds.

Knob	Parameter	Does
HARM	Engine Parameter 1	Detune between the two oscillators, following musical intervals
TIMBRE	Engine Parameter 2	Table position of oscillator 1
MORPH	Engine Parameter 3	Table position of oscillator 2

WAVETABLE SWARM

A supersaw-style swarm of wavetable voices.

Knob	Parameter	Does
HARM	Engine Parameter 1	Detune spread
TIMBRE	Engine Parameter 2	Table position (browse as in WAVETABLE)
MORPH	Engine Parameter 3	Voice count

5.5 Acid

ACID 1

The original 303-style acid bass. Frozen and unchanged for compatibility.

Knob	Parameter	Does
HARM	Engine Parameter 1	Overdrive
TIMBRE	Engine Parameter 2	Waveform, saw to square
MORPH	Engine Parameter 3	Accent

ACID 2

Our faithful 303 rework. A variable-shape oscillator through a finely-tuned gritty filter, with four distortion zones and an optional Devilfish mode. 4-voice polyphonic (a 5th note steals the oldest).

Knob	Parameter	Does
HARM	Engine Parameter 1	Distortion zone (sweeps; the LED ring shows the zone)
TIMBRE	Engine Parameter 2	Waveform, saw to square
MORPH	Engine Parameter 3	Accent amount

Zone	Sound	Good for
SCREAMER	Smooth classic acid overdrive	Your default; keep low for clean lines
VALVE	Warm tube saturation, never harsh	Melodic lines, slower acid
MIXER	Overdriven desk channel	The acid-techno scream: high reso, sweep the cutoff
OCTAFUZZ	Octave-up fuzz	Basslines that scream an octave above

Zones cross-fade, so park between two for hybrids. To glide, use a CV/Gate pair and change the pitch while the gate is high. On a mono MIDI part, send legato notes. Accents come from note velocity or the Amp CV.

> **NOTE** Turn on Devilfish by setting devfsh 1 in options.txt (Chapter 8) for a sub-oscillator, fatter resonance and hotter drive.

5.6 Keys, Strings and Physical Models

STRING 2

A string-machine voice: strings, brass, organ and choir, with an ensemble and phaser. Supports one-finger smart chords.

Knob	Parameter	Does
HARM	Engine Parameter 1	Registration
TIMBRE	Engine Parameter 2	Ensemble
MORPH	Engine Parameter 3	Phaser

> **NOTE** Smart chord: turn it on in the options ring (LED 10). The lowest note you hold sets the key. A single note then plays a diatonic chord. The encoder feature stop picks the scale. MORPH sets the voicing, from octave and fifth up to a full spread.

PIANOZ

A rompler voice: house piano, Rhodes electric piano and drawbar organ, streamed from the SD card.

Knob	Parameter	Does
HARM	Engine Parameter 1	Instrument: piano (LEDs 1 to 3), Rhodes (LEDs 4 to 7), organ (LEDs 8 to 10)
TIMBRE	Engine Parameter 2	Octave stack: right fades an octave up (bright), left a sub octave (deep). On organ, left is the 16′ sub bar, right the sparkle bars.
MORPH	Engine Parameter 3	Lo-fi crush: clean at 0, vintage grit by a quarter turn, full crush at max

> **NOTE** PIANOZ streams HOUSEKEYS.PCM and RHODES.PCM from the System/ folder on the SD card. The Coral Factory package includes them (Section 8.1).

> **NOTE** PIANOZ also supports smart chord (options ring, LED 10), with the same 9-scale picker as String 2.

STRING

Physical-model plucked and bowed string / resonator.

Knob	Parameter	Does
HARM	Engine Parameter 1	Material (amount of inharmonicity)
TIMBRE	Engine Parameter 2	Excitation brightness and dust density
MORPH	Engine Parameter 3	Decay time (energy absorption)

MODAL

Physical-model resonant bodies and mallets.

Knob	Parameter	Does
HARM	Engine Parameter 1	Structure
TIMBRE	Engine Parameter 2	Excitation brightness
MORPH	Engine Parameter 3	Decay / damping

5.7 Noise and Drums

PARTICLE NOISE

Granular particle and noise clouds.

Knob	Parameter	Does
HARM	Engine Parameter 1	Frequency spread
TIMBRE	Engine Parameter 2	Density
MORPH	Engine Parameter 3	Resonance

FILTERED NOISE

Tuned, filtered noise.

Knob	Parameter	Does
HARM	Engine Parameter 1	Pitch / mode
TIMBRE	Engine Parameter 2	Cutoff
MORPH	Engine Parameter 3	Resonance

SNARE / HI-HAT

A combined snare and hi-hat voice.

Knob	Parameter	Does
HARM	Engine Parameter 1	Harmonic / noise balance
TIMBRE	Engine Parameter 2	Body / hi-hat high-pass
MORPH	Engine Parameter 3	Snare to hi-hat crossfade

HI-HAT

Analog-style hi-hat: two models blended.

Knob	Parameter	Does
HARM	Engine Parameter 1	Metallic / filtered-noise balance
TIMBRE	Engine Parameter 2	High-pass cutoff
MORPH	Engine Parameter 3	Crossfade between the two models

SNARE

Analog-style snare: two models blended.

Knob	Parameter	Does
HARM	Engine Parameter 1	Harmonic / noisy balance
TIMBRE	Engine Parameter 2	Body / noise components
MORPH	Engine Parameter 3	Crossfade between the two models

BASSDRUM

Analog-style kick: two models blended.

Knob	Parameter	Does
HARM	Engine Parameter 1	Attack sharpness and overdrive
TIMBRE	Engine Parameter 2	Brightness

Knob	Parameter	Does
MORPH	Engine Parameter 3	Crossfade between the two models

On the drum engines you can send CV modulation to any parameter. Use the velocity or mod outputs of a sequencer, for example. Over MIDI you can play with velocity. The Mod decay (SHIFT) changes the internal envelope decay. The Amp decay refines the amplitude.

5.8 Generative Drone

STITCHER

A generative drone looper, and the one engine that plays by its own rules. Stitcher holds a 2 second window of a sample in memory and chops it into three independent loops. Every re-roll rebuilds each loop as a fresh arrangement of slices. The slices use mixed musical lengths, shuffled order, repeats and reversals. Each slice has its own level, pan and pitch. Slices are played like an oscillator rather than gated, so the result flows instead of stuttering.

Read this before you reach for the knobs. On Stitcher the OCT knob is not pitch, and HARM, TIMBRE and MORPH are not the usual engine parameters. They are the levels of the three loops.

Control	On STITCHER
OCT · the big knob	The main control, and not pitch on this engine. Any meaningful move re-rolls the whole arrangement, and its position sets the density. MIDI CC 6 drives it too.
HARM / TIMBRE / MORPH	Level of loop 1 / 2 / 3. The three loops are roles rather than copies: percussive, spacey and smeared, so they behave like three different players.
MOD ATK	Bipolar rhythm to texture morph. Centre is neutral. One way gives short slices, rests and a wide pitch spread. The other gives long overlapping slices that sit with the root.
MOD DR	Its normal Coral mod-decay job.
Keys / CV note	Pitch the material. Stitcher is a single-voice texture engine.

Picking a sample

The first encoder click opens the sample browser, shown as a yellow arc. Turn to scrub through the WAV files in your Stitcher folder, then click to exit. Position 0 is the built-in generator, so the engine makes sound even with an empty folder. Your choice is remembered across power cycles.

> **NOTE** Coral creates the **Stitcher** folder at the root of the SD card on boot, so there is nothing to set up. Copy your WAV files straight into it. That folder at the card root is the only place Stitcher reads, and it does not look inside subfolders. A long file is fine. Each re-roll can take its 2 second window from anywhere in the file. One sample goes a long way.

> **TIP** The Coral Factory package puts a set of sounds in the Stitcher folder (Section 8.1). You can add your own. Stitcher also plays with an empty folder, using its built-in generator at position 0.

The FX ring

A second encoder press opens the Stitcher FX ring: three live toggles you can flip while it plays. A slot at full brightness is on, dim is off. The settings are saved on the device, so they survive a reboot. Hold about 1 second to exit, or leave it and it closes after about 4 seconds.

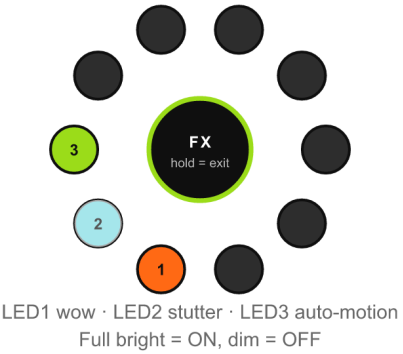


Figure 5.1. Stitcher FX ring: LED1 tape wow and flutter, LED2 stutter and re-trigger bursts, LED3 auto-motion. Full brightness is on, dim is off.

LED	FX	What it does
1	**Tape wow and flutter**	Slow pitch and speed wobble, like tape.
2	**Stutter**	Re-trigger bursts across the arrangement.
3	**Auto-motion**	The big knob turns itself, so the arrangement keeps re-rolling on its own.

TIP For a hands-off generative patch, pick a sample. Set the three loop levels to taste. Turn on auto-motion (LED 3). Coral then rebuilds the arrangement by itself.

5.9 Sampler

WAVPLAYER

Plays your .wav samples from the SD card. Any bit depth, sample rate, length, mono or stereo. One-shot playback, polyphonic.

Knob	Parameter	Does
HARM	Engine Parameter 1	Overdrive
TIMBRE	Engine Parameter 2	Sample selection
MORPH	Engine Parameter 3	Folder selection

Organize samples in up to 10 folders with 32 samples each. Coral ignores empty folders. It orders folders and samples alphabetically. You can modulate the sample and folder selection. Coral supports 16, 24, 32 and 32-float bit depths, at 44.1 or 48 kHz. Samples pitch up 18 semitones and down 24, by a change of playback speed. There is no time-stretching.

CAUTION Heavy sampler use can still produce artifacts under load. Card speed matters most. Use a U3 card. Keep samples at 48 kHz 16-bit. Use fewer voices on the sampler at once.

6 SOUND EFFECTS

There are three effects: Space (reverb), Chorus (ensemble) and Delay. All are send effects common to all voices; the amount depends on each voice level and its send amount.

> **NOTE** The three send amounts and the HP filter cutoff stay with the part when you change its engine. See Section 2.4. Your effects balance survives an engine switch on stage.

6.1 Space

The Space engine is not only a reverb but also improves the stereo image of the dry signal. Send amount is set per voice with SHIFT + FILTER. The size and low-pass of the reverb are set over MIDI (CC 40 and 41).

6.2 Chorus

The chorus is applied per voice. Send amount is set with SHIFT + MORPH when the high-pass filter option is off. Turn on the high-pass filter option and SHIFT + MORPH becomes the per-voice HP cutoff. The chorus send is then not available.

6.3 Delay

The delay is synced to the tempo. Turn on Delay in the options ring (LED 8). SHIFT + TIMBRE then sets the delay send of each voice. It replaces the noise level. Over MIDI, the send is CC 27 on the MIDI channel of the part.

Tempo

The delay follows the MIDI clock on the MIDI input. With no clock, it runs at 120 BPM. After a tempo change, it needs a few seconds to settle. When the clock stops, the delay keeps the last tempo.

Delay settings over MIDI

The other delay settings have no panel control. You set them with MIDI CCs only. They work on any MIDI channel, and the preset saves them.

CC	Setting
44	Time, as a note value (see the table below)
45	Feedback
46	Stereo spread: the left repeats are up to 25% longer than the right repeats
47	Low-pass filter on the repeats (a lower value gives a darker sound)

CC 44 values. One beat is a quarter note.

Value	Delay time	Value	Delay time	Value	Delay time
0 to 8	Free, very short (like a flanger)	47	1/16 triplet	89	Dotted 1/8
13	1/64 triplet	55	1/16	97	1/4
21	1/64	64	1/8 triplet	106	1/2 triplet
30	1/32 triplet	72	1/8	114	Dotted 1/4
38	1/32	80	1/4 triplet	123	1/2

Defaults: time 1/8, feedback about 40%, spread 20%, low-pass 90%.

> **NOTE** The longest delay is 1.2 seconds. A 1/2 note needs a tempo of 100 BPM or more. A dotted 1/4 needs 75 BPM or more. Coral cuts a longer time to 1.2 seconds.

6.4 High-Pass Filter

As an alternative to the chorus, a per-voice stereo high-pass filter can help Coral sit in a mix. Turn it on in the options ring (LED 7); SHIFT + MORPH then controls the HP cutoff. Over MIDI the HP cutoff is CC 20.

7 MIDI IMPLEMENTATION

With MIDI CC messages you can modulate any parameter inside Coral. Messages are filtered by MIDI channel and applied to the part on that channel. CC is unipolar and overrides the knob value.

7.1 Per-Part CC Table

Send these on the part's MIDI channel. The CC overrides the current knob position.

CC	Parameter
1	Filter cutoff
2	Filter resonance
3	Filter-env amount
4	Level
5	Pan
6	Octave (on STITCHER: the big knob)
7	Glide amount
8	Glide mode
11	Engine Param 1 (HARM)
12	Engine Param 2 (TIMBRE)
13	Engine Param 3 (MORPH)
14	Amp attack
15	Amp decay / release
16	Mod attack
17	Mod decay / release
18	Sustain
19	Noise level
20	HP filter cutoff
25	Chorus send (CPU-free when 0)
26	Space (reverb) send
27	Delay send
30	Engine select (value 0 to 9 = engine slot 1 to 10; a value above 9 = slot 10)
64	Sustain pedal (hold notes / chords)

7.2 Global FX CC Table

These work on any MIDI channel. They are the only way to reach the reverb and delay engine parameters, which have no panel knob.

CC	Parameter
40	Reverb size / length
41	Reverb low-pass
44	Delay time (the timing follows the MIDI clock, see Section 6.3)
45	Delay feedback
46	Delay spread (ping-pong)

CC	Parameter
47	Delay low-pass

7.3 CC and the Panel Pots

A CC takes over the parameter it addresses. The panel pot must catch up before it takes control again. As you turn, the pot and the value meet in the middle. Section 2.2 covers this. It also covers the noccxt setting, which keeps the pots live under a stream of CCs.

This works for every CC in the table above that has a pot. It includes the Space (reverb) and Delay sends. You can automate the sends from your DAW, then pick them up on the panel.

7.4 Program Change

Program Change messages with a value of 1 to 10, received on any MIDI channel, load presets 1 to 10. You can change this in the options ring (Program Change slot): accept all, ignore all, or accept only on an active part channel.

7.5 Pitch Bend

Pitch Bend is applied to the part whose MIDI channel matches the message. The default range is ±2 semitones. You can set it from 1 to 48 with the pbrang option (Chapter 8).

7.6 Glide

Glide is a smooth transition of a voice's pitch from one note to the next. In polyphonic mode different notes can trigger different voices, so glide can be unpredictable. Use a single-voice (mono) part for a predictable glide. Glide time is CC 7. Glide mode is CC 8: 0 selects glide always, 1 to 127 selects glide on legato only.

7.7 Sustain Pedal

CC 64 acts as a sustain / damper pedal, holding notes and chords while it is down.

7.8 Velocity and Pressure

Note velocity is mapped to level. Channel and polyphonic pressure (aftertouch) modulate the filter cutoff and level for immediate playability.

7.9 MPE

Coral supports MPE controllers. MPE lets you control pitch bend (horizontal movement), pressure (aftertouch) and slide (vertical movement) per note. Aftertouch drives the filter cutoff and the volume. You can assign slide, or any other expressive dimension, to any CC parameter of Coral. Turn MPE on in the options ring (LED 4).

> **NOTE** In MPE mode you cannot set the MIDI channels of the voices. The MPE engine sets them for you.

8 SD CARD AND OPTIONS

The Micro SD card holds the factory content. Always keep it in the slot. Heavy sampler use needs a Class 10 card with a read speed of 70 MB/s or more. To use a larger or faster card, format it as FAT32. Then set it up with the Coral Factory package (Section 8.1).

8.1 The Coral Factory Package and the Card Layout

Every new or empty card needs the Coral Factory package.

1. Download the Coral Factory package.
2. Unzip it.
3. Copy everything inside it to the root of your SD card. If your computer asks, merge the folders. If it asks to replace CALIB.bin, choose Skip (or Keep both).

CAUTION Your card may already have these folders. That is not enough: the package puts the files inside them (samples, DX7 patches, Stitcher sounds, piano data). Without them some engines are silent. options.txt is made by Coral itself the first time it starts. Never replace your System/CALIB.bin: it holds your own Coral's factory calibration, and a copy from another Coral detunes it.

This is the layout of a correctly set-up card. A folder marked with an open arrow shows what goes inside it. The files HOUSEKEYS.PCM, LUT.bin, options.txt and RHODES.PCM go inside the System folder, not at the root of the card.

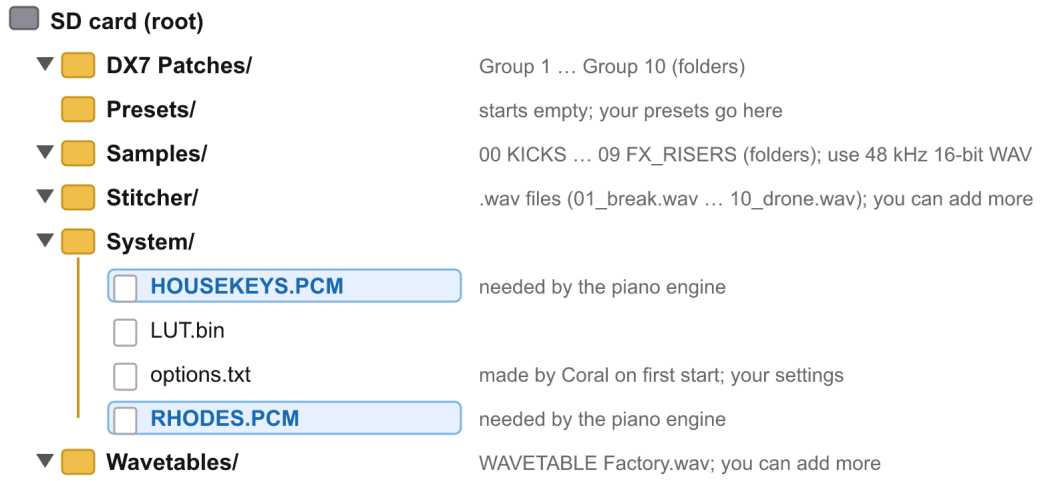


Figure 8.1. SD card layout. At the root: DX7 Patches, Presets, Samples, Stitcher, System and Wavetables. Inside System: HOUSEKEYS.PCM, LUT.bin, options.txt and RHODES.PCM.

Folder	Contents
DX7 Patches	.syx banks for FMX7, in the folders Group 1 to Group 10 (one bank each).
Presets	Starts empty. Your presets go here as Preset N.preset files (the full 8-voice configuration).
Samples	Samples for WAVPLAYER, in the folders 00 KICKS to 09 FX_RISERS. Use 48 kHz 16-bit WAV.
Stitcher	.wav files for the STITCHER engine. The package includes a set, and you can add more. Put them directly in this folder: Coral does not read subfolders.

Folder	Contents
System	HOUSEKEYS.PCM and RHODES.PCM: needed by the piano engine (PIANOZ). LUT.bin. options.txt: made by Coral on first start, your settings (Section 8.3). Your card also holds CALIB.bin, the factory calibration of your own Coral. It is not in the package. Never replace it with a copy from another Coral. Calibrating at home writes a new one (Chapter 10).
Wavetables	WAVETABLE Factory.wav for the wavetable engines. You can add more.

8.2 Wavetables

The Wavetables folder should contain at least one .wav wavetable. As long as the naming is right, Coral loads it for the wavetable engines. Make your own with the Waveform Generator (Chapter 9).

8.3 options.txt

options.txt is a plain-text file in the System folder, created on first boot. Every setting Coral has lives here, one per line: a six-letter key, a space, then a value (usually 1 for on and 0 for off). Order does not matter. You can edit it on a computer with any text editor.

The SD card owns your settings. Coral reads them from the card at boot. Only you change them, in this file or on the options ring. Updating the firmware never alters them.

Which settings the file owns, and which the ring owns

Most of these settings can also be changed on the Coral itself, on the options ring (Chapter 2). The two work together, with one simple rule.

- **Ring options are in the file too.** Whenever you toggle one on the ring, Coral refreshes options.txt to match. While you are playing, the ring is in charge.
- **Your edits win.** If you edit options.txt on a computer, your version is used on the next boot. So the file is always a safe way to set things up.
- **Some settings are file only.** vumetr, ledalw, extpre, obvspr, pbrang, noccxt, devfsh and engcol have no place on the ring. You set them here only.

> **TIP** Upgrading from firmware 2.6? Your old options.txt works exactly as it did. polycv, encpri and cvpart are normal keys again, so a 2.6 file is read verbatim.

The full key list

Key	Setting	Values
polycv	PolyCV voice / trigger mode	0 unison · 1 poly CV · 2 chord hold · 3 chord arp · 4 and 5 echo
encpri	Encoder part priority	1 = encoder turn selects part / voice · 0 = selects engine (default)
cvpart	CV part to engine	1 = Part CV selects the engine · 0 = selects the part (default)
mpeena	MPE	1 = MPE in and out · 0 = off (default)
pcigno	Program Change	0 accept all · 1 ignore all · 2 accept only on an active part channel
rngenc	Randomizer	1 = hold the encoder to randomize · 0 = off (default)
hpfilt	High-pass filter	1 = SHIFT + MORPH is HP cutoff · 0 = chorus send (default)
delena	Delay FX	1 = SHIFT + TIMBRE is delay send · 0 = noise level (default)
strchd	Smart chord, String 2	1 = on · 0 = off (default)
pnochd	Smart chord, PIANOZ	1 = on · 0 = off (default)
vumetr	VU meter (file only)	1 = master level on LEDs 9 and 10 · 0 = off (default)
ledalw	Selection LED (file only)	1 = always show the selected part / voice · 0 = off (default)
extpre	Extended presets (file only)	1 = 100 presets (10 groups of 10) · 0 = 10 presets (default)
obvspr	OB-VA spread (file only)	1 = per-voice analog drift on OB-VA · 0 = off (default)
pbrang	Pitch-bend range (file only)	Semitones, 1 to 48 · default 2
noccxt	CC arms catch-up (file only)	0 = an incoming CC arms pot catch-up (default) · 1 = CCs only change the sound and the pots stay live
devfsh	ACID 2 Devilfish (file only)	1 = sub-oscillator, fatter resonance, hotter drive · 0 = off (default)

Key	Setting	Values
engcol	Engine indicator colour (file only)	1 = colour the engine indicator per engine · 0 = one flat magenta (default)

NOTE Smart chord (strchd and pnochd) is simply on or off. The musical scale is chosen separately, on the encoder feature stop (Section 5.6).

TIP The file explains itself. Every setting line ends with a short note on what that setting does. You can edit the card without this manual next to you. Coral fills the notes in for you if your file has none.

NOTE Anything you write yourself is kept. Your own comments, blank lines and any line Coral does not recognise come back exactly as you typed them. Coral rewrites the file only when something has changed.

8.4 Presets on the Card

Presets are stored as Preset N.preset files in the Presets folder. With the extpre option on you get 100 presets in 10 groups of 10. You can back up, share and edit these files with the Preset Manager (Chapter 9).

9 COMPANION WEB APPS

Three free web apps support Coral, Firmware Updater, Preset Manager and Waveform. All run in your browser. The Firmware Updater builds and flashes firmware, the Preset Manager builds and manages presets, and the Waveform Generator makes wavetables.

9.1 Firmware Updater

The Firmware Updater does three things. You choose your 10 engines or less. The OXI build server compiles a custom firmware. You flash it to Coral over USB. Use Chrome or Edge, which support WebUSB.

Step 1 · Put Coral in update mode

Hold the BOOT button on the Daisy board inside the module. Connect Coral to the computer by USB. It appears as a DFU device. Power it from USB only, with nothing else connected.

Step 2 · Connect

Click Connect Coral and select the DFU device. The app can read back and show the firmware currently installed.

Step 3 · Build or choose firmware

- Start from a shipped layout in the preset menu, or pick your own 10 engines in the grid. The chosen engines fill the 10 slots in order.
- Watch the Flash and RAM meters. Green means it will build. Some engines are large (FMX7, PIANOZ and STITCHER especially), so a combination of several heavy engines may not fit. The meters tell you before you build.
- Leave WAVPLAYER out of your 10 and Coral drops the sampler from the build. Its memory comes back. That leaves room for heavier engines.
- Your settings are never touched. Flashing changes the firmware only: your options live on the SD card and are read from there on the next boot.
- Click Build firmware. Every build compiles a fresh firmware on the server. You can also save your 10-engine layout as a .json file to reload or share. You can print a one-page cheat sheet for your build.

> **NOTE** Instead of building, you can switch to the Upload .bin tab and flash a .bin you already have.

Step 4 · Flash

Do not unplug the module while it flashes. Click Flash to Coral and wait. Coral restarts on the new firmware when it finishes.

> **CAUTION** Flashing replaces the firmware, not your SD card. Your presets, samples and wavetables stay on the card. If you added an engine that needs files on the card, make sure they are there. PIANOZ needs HOUSEKEYS.PCM and RHODES.PCM in System/. FMX7 needs its DX7 banks (Section 8.1).

10 CALIBRATION

Calibrate the CV/Oct tracking if pitch drifts. You need a well-calibrated ±1 V source.

1. Turn Amp Atk, Amp Dr, Sust, Mod Atk, Mod Dr and F.Reso fully clockwise, and the F.Env trimmer fully counter-clockwise.
2. Unplug all patch cables from the module.
3. Hold the encoder for 5 seconds until the ring shows the green pattern.
4. Tap the encoder; the ring turns red.
5. Patch the OCT CV input and send -1 V from a well-calibrated source. (On the OXI One, C1 outputs -1 V.)
6. Tap the encoder once; the ring shows red and amber.
7. Send +1 V to the OCT input and tap the encoder again. (On the OXI One, C3 outputs +1 V.)
8. Tap the encoder again. If it worked, the green confirmation animation appears.

> **NOTE** Depending on your voltage source, -1 V may correspond to a different MIDI note.

10.1 Fix Tuning If You Only Use MIDI

This fix is for MIDI only. Use it if your Coral plays out of tune after installing 3.0. In that case its calibration file was replaced by the one in the old Factory package. You can fix it in one minute, with no cables.

1. Unplug every cable from the Coral's CV inputs. Leave MIDI connected if you want.
2. Turn Amp Atk, Amp Dr, Sust, Mod Atk, Mod Dr and F.Reso fully clockwise, and the F.Env trimmer fully counter-clockwise.
3. Hold the encoder down for about 5 seconds, until the LEDs turn cyan and white.
4. Click the encoder 3 times, waiting a second between clicks. The LEDs change colour after each click.
5. After the third click the LEDs light up. This is normal: the Coral has already saved the fix.
6. Turn the Coral off and on again.

Your Coral now plays in tune from MIDI. Set your knobs back where you like them.

> **NOTE** This fix does not calibrate the OCT CV input. If you play Coral from CV/Oct, use the full calibration above with a well-calibrated ±1 V source.

A APPENDIX: ENGINE ROSTER

All 27 engines and what HARM, TIMBRE and MORPH do in each. Pick any 10 in the Firmware Updater.

Engine	HARM	TIMBRE	MORPH
VIRTUAL ANALOG	Osc detune	Square shape	Saw shape
WAVESHAPE	Waveform	Fold amount	Asymmetry
2-OP FM	Ratio	FM depth	Feedback
WAVETABLE	Z axis	X axis	Y axis
DUAL WAVETABLE	Detune	Table pos, osc 1	Table pos, osc 2
WAVETABLE SWARM	Detune spread	Table position	Voice count
MDO	Detune	Saw to square	Pulse width
MDO2	Detune spread	Waveform	Voice count
ADDITIVE	Harmonic center	Fold amount	Diode drive
ACID 1	Overdrive	Waveform (saw to square)	Accent
ACID 2	Distortion zone	Waveform (saw to square)	Accent
3-VCO ANALOG	Osc 2 detune	Waveform	Osc 3 detune
SH-101	Sub tuning	Waveform and PWM	Sub level
OB-VA	Detune / spread	VCO combo (upper = unison)	Filter slope 4 to 2 pole
STRING 2	Registration	Ensemble	Phaser
FMX7 (DX7)	Patch (in bank)	FM brightness	Juno chorus
PIANOZ	Instrument	Octave stack	Lo-fi crush
STRING	Material	Excite brightness	Decay
MODAL	Structure	Excite brightness	Decay / damping
PARTICLE NOISE	Frequency spread	Density	Resonance
FILTERED NOISE	Pitch / mode	Cutoff	Resonance
STITCHER	Loop 1 level	Loop 2 level	Loop 3 level
SNARE / HI-HAT	Harmonic / noise	Body / hat HP	Snare to hi-hat
HI-HAT	Metallic / noise	HP cutoff	Model blend
SNARE	Harmonic / noise	Body / noise	Model blend
BASSDRUM	Attack / drive	Brightness	Model blend
WAVPLAYER	Overdrive	Sample	Folder

NOTE STITCHER is the exception: its three knobs are loop levels, and its main control is OCT. See Section 5.8.

B APPENDIX: TROUBLESHOOTING

Symptom	What to do
MIDI stops responding until I power cycle, but the sound keeps playing	Update to firmware 3.0. A single corrupted MIDI byte could lock up the MIDI input on older firmware. A worn cable or connector produces one sooner or later. Audio carried on, so the module looked frozen while it still droned. Firmware 3.0 recovers from a corrupted byte at once. If it continues after you update, try a different MIDI cable.
Coral freezes and only a power cycle brings it back	Check the SD card. A card can refuse writes when it is write-protected, worn out, or has a damaged filesystem. Older firmware then kept retrying the save of your settings, and stopped responding while it did. Firmware 3.0 gives up after a few tries instead of hanging. Make sure the card is not write-protected. Copy your content to a fresh card if you think it is failing.
No sound at all	Make sure the Micro SD card is in the slot. Make sure the part you play is on the MIDI channel you send. Make sure the voice level is up. Hold the encoder for 5 seconds to return Coral to its default state (Section 3.4).
One engine is silent while the others play	That engine may need files on the SD card. PIANOZ needs HOUSEKEYS.PCM and RHODES.PCM in System/, and FMX7 needs its DX7 banks. Copy the Coral Factory package to the card (Section 8.1). Seeing the folders on the card is not enough: the files must be inside them.
Some of my samples do not show up, or Coral takes a long time to start	Update to firmware 3.0. Older firmware read a large piece of every WAV while it scanned the card. That made start-up slow on a full card. It could also leave files out of the list. Firmware 3.0 reads only what it needs. Start-up is much quicker, and it finds every playable file.
Clicks or artifacts from the sampler	Crackle or dropouts with several sampler voices at once is a different thing. That is card speed. Use a fast card (U3). Convert samples to 48 kHz 16-bit. Use fewer voices on the sampler at once (Section 5.9).
A pot does nothing until I turn it a long way	It is catching up with a value set over MIDI, or by a part or preset change: keep turning and the value comes to meet you. Section 2.2 explains this, and noccxt can stop CCs arming catch-up at all.
Pitch is off when playing from CV	Calibrate the CV/Oct input (Chapter 10).
My Stitcher samples disappeared	Stitcher reads only the Stitcher folder at the root of the card. Move your WAV files there (Section 5.8).

Changelog

What is new since manual v2.1 (firmware 3.0):

- New engine model: 27 engines, choose any 10 with the web Firmware Updater.
- New engines: FMX7 (6-op DX7 with .syx banks), ACID 2 (four distortion zones, Devilfish), OB-VA and STRING 2 (smart chord). Also PIANOZ (rompler keys), STITCHER (generative drone looper), SH-101, Dual Wavetable, Wavetable Swarm, MDO2, Modal, Particle Noise and Filtered Noise.
- On-device options ring: set the performance options without a computer. Coral mirrors them into options.txt. A 2.6 options.txt still works as it is.
- Encoder click cycle with per-engine feature stops (FMX7 bank, wavetable selection, Stitcher sample and FX ring).
- PolyCV trigger modes 0 to 5, smart chord, randomizer.

- Drone mode: a setting of each part, from the options ring. The last notes you play keeps sounding. The preset saves it.
- New MIDI: HP filter cutoff (CC 20), sustain pedal (CC 64), MPE, velocity and pressure.
- CC catch-up: an external CC now arms panel catch-up properly, including the Delay send. noccxt keeps the pots live instead, if you prefer.
- Per-preset wavetable memory; up to 100 presets with extpre.
- Per-engine tone memory: each engine on a part keeps its own tone settings. The reverb, delay and chorus sends stay with the part, so an engine change never disturbs your FX mix.
- FMX7: the DX7 bank and patch are now per part. Several FMX7 parts can play different sounds at once. The preset saves the bank.
- Faster start-up with a full sample card, and no more samples missing from the list.
- Sampler: samples play to their true end and retriggering no longer clicks.
- Flashing new firmware never changes your settings: the SD card owns them.
- New companion apps: Firmware Updater, Preset Manager, Waveform Generator.
- Reliability: a corrupted MIDI byte can no longer lock up the MIDI input. A failing SD card can no longer hang the module while it saves your settings (see Appendix B).

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DECLARATION OF CONFORMITY

This product complies with the requirements of **European Directive 2014/53/EU and Directive 2011/65/EU.** The full text of the EU declaration of conformity is available at www.oxiinstruments.com.

Environmental issues: This symbol (crossed-out wheeled bin) indicates that this product should not be treated as domestic waste. At the end of its useful life it must be taken to a collection point for the recycling of electrical appliances. Please inform yourself about local, state and European regulations regarding proper disposal and recycling of products containing lead, batteries and plastics.

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Refund policy (European Union only)

The consumer has 14 days from acquisition to return the product for a full refund. The product must be in the same state and with all original content. Shipping costs are paid by the customer.

Warranty

OXI Instruments warrants the hardware and accessories against defects in materials and workmanship for two years from the date of original purchase. Unless proven otherwise, breaches of conformity within six months of delivery are presumed to have existed at delivery. The warranty does not cover:

- Misuse of OXI Coral, including use under extreme conditions or incorrect use.
- Improper handling of the product.
- Normal wear and tear, or damage caused by accident or abuse.
- Malfunction due to accessories not authorized by OXI Instruments.

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